

Chapter 7 — Exercises

Multiple-Choice Questions

Working with text data

How to Use These Slides

- **12 multiple-choice** questions on Chapter 7 concepts
- Pick **one** answer (A–D) for each question
- The **answer key** with brief explanations is at the end
- Try to answer before revealing the solutions

Topics: bag-of-words, CountVectorizer, min_df & stopwords, tf-idf, n-grams, stemming/lemmatization, and LDA.

Q1. The bag-of-words representation discards:

- A. the vocabulary
- B. word order
- C. the word counts
- D. the documents themselves

Q2. `CountVectorizer` produces:

- A. a sparse matrix of word counts
- B. a dense image
- C. a single integer per document
- D. a list of sentences

Q3. Setting `min_df=5` means a token is kept only if it:

- A. appears at most 5 times total
- B. is exactly 5 characters long
- C. appears in at least 5 documents
- D. is among the top 5 words

Q4. Stopwords are:

- A. rare, highly informative words
- B. words unique to one document
- C. misspelled words
- D. very frequent words that carry little signal

Q5. tf-idf gives a high weight to a word that is:

- A. frequent in one document but rare across the corpus
- B. frequent in every document
- C. absent from the corpus
- D. always a stopword

Q6. Using n-grams (e.g. bigrams) lets the model capture:

- A. document length only
- B. some word order and context
- C. fewer features
- D. the target labels directly

Q7. The feature "not worth" is an example of:

- A. a stopword
- B. a stemmed token
- C. a bigram that captures context
- D. a topic

Q8. Compared to stemming, lemmatization:

- A. is always faster
- B. ignores the dictionary
- C. only trims common suffixes
- D. uses a dictionary and the word's role for an accurate base form

Q9. Latent Dirichlet Allocation (LDA) is used for:

- A. unsupervised topic modeling
- B. supervised classification
- C. feature scaling
- D. cross-validation

Q10. The words tf-idf marks as "important" are:

- A. always predictive of the label
- B. distinctive, but not necessarily predictive of the label
- C. the rarest single-document words
- D. exactly the stopwords

Q11. For high-dimensional sparse text features, a good model is:

- A. a deep decision tree
- B. k-nearest neighbors
- C. a linear model like logistic regression
- D. DBSCAN

Q12. Adding trigrams to the vocabulary tends to cause:

- A. fewer features
- B. a guaranteed drop in accuracy
- C. no change at all
- D. an explosion in the number of features

Answer Key

Solutions & Explanations

Answers 1–6

Q	Ans	Why
1	B	Bag-of-words keeps counts but discards word order
2	A	Output is a sparse count matrix
3	C	<code>min_df=5</code> → token must appear in ≥5 documents
4	D	Stopwords = very frequent , low-signal words
5	A	High tf-idf: frequent in a doc, rare in the corpus
6	B	n-grams recover word order / context

Answers 7–12

Q	Ans	Why
7	C	"not worth" is a bigram that captures context
8	D	Lemmatization uses a dictionary + word role
9	A	LDA is unsupervised topic modeling
10	B	tf-idf is unsupervised → distinctive ≠ predictive
11	C	High-dim sparse text → linear models work best
12	D	More n-grams → feature explosion

Well done!

Review Chapter 7 for any you missed